

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 2

Adaptive Classified

Chapter 09 · Laws of Logarithms

Questions

Student book · questions with writing space

- 46 pages
- Past papers up to January 2026

PURE 2 | TOPIC 09

Laws of Logarithms

Your classified route

20

QUESTIONS

148

MARKS

4

FAMILIES

5

VARIANTS

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

TOPIC 09 | FAMILY 01

Change Of Base

WHAT THIS FAMILY ASKS

Recognise the change of base demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**1 classified questions**

7 total marks

10. Given $a = \log_2 3$

(i) write, in simplest form, in terms of a ,

(a) $\log_2 9$

(b) $\log_2 \left(\frac{\sqrt{3}}{16} \right)$

(3)

Adaptive unit 1 · WMA12-01_2022_Oct_Q10

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10. Given $a = \log_2 3$

(ii) Solve

$$3^x \times 2^{x+4} = 6$$

giving your answer, in simplest form, in terms of a .

(4)

Adaptive unit 2 · WMA12-01_2022_Oct_Q10

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TOPIC 09 | FAMILY 02

Log Definition

WHAT THIS FAMILY ASKS

Recognise the log definition demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**1 classified questions**

7 total marks

9. (a) Sketch the curve with equation

$$y = 3 \times 4^x$$

showing the coordinates of any points of intersection with the coordinate axes.

(2)

Adaptive unit 1 · WMA12-01_2020_Jan_Q09

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The curve with equation $y = 6^{1-x}$ meets the curve with equation $y = 3 \times 4^x$ at the point P .

(b) Show that the x coordinate of P is $\frac{\log_{10} 2}{\log_{10} 24}$ (5)

Adaptive unit 2 · WMA12-01_2020_Jan_Q09

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TOPIC 09 | FAMILY 03

Log Laws

WHAT THIS FAMILY ASKS

Recognise the log laws demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE**V01** core family move**3 classified questions**

20 total marks

7. Given $\log_a b = k$, find, in simplest form in terms of k ,

(i) $\log_a \left(\frac{\sqrt{a}}{b} \right)$

(2)

Adaptive unit 1 · WMA12-01_2019_Oct_Q07

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7. Given $\log_a b = k$, find, in simplest form in terms of k ,

(7)

(ii) $\frac{\log_a a^2b}{\log_a b^3}$

(2)

Adaptive unit 2 · WMA12-01_2019_Oct_Q07

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7. Given $\log_a b = k$, find, in simplest form in terms of k ,

(7)

(iii) $\sum_{n=1}^{50} (k + \log_a b^n)$

(3)

Adaptive unit 3 · WMA12-01_2019_Oct_Q07

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4. The weight of a baby mammal is monitored over a 16-month period.

The weight of the mammal, w kg, is given by

$$w = \log_a(t + 5) - \log_a 4 \quad 2 \leq t \leq 18$$

where t is the age of the mammal in months and a is a constant.

Given that the weight of the mammal was 10 kg when $t = 3$

(a) show that $a = 1.072$ correct to 3 decimal places.

(3)

Adaptive unit 1 · WMA12-01_2022_Oct_Q04

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4. The weight of a baby mammal is monitored over a 16-month period.

The weight of the mammal, w kg, is given by

$$w = \log_a(t + 5) - \log_a 4 \quad 2 \leq t \leq 18$$

where t is the age of the mammal in months and a is a constant.

Given that the weight of the mammal was 10 kg when $t = 3$

Using $a = 1.072$

(b) find an equation for t in terms of w

(3)

Adaptive unit 2 · WMA12-01_2022_Oct_Q04

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4. The weight of a baby mammal is monitored over a 16-month period.

The weight of the mammal, w kg, is given by

$$w = \log_a(t + 5) - \log_a 4 \quad 2 \leq t \leq 18$$

where t is the age of the mammal in months and a is a constant.

Given that the weight of the mammal was 10 kg when $t = 3$

(c) find the value of t when $w = 15$, giving your answer to 3 significant figures.

(2)

Adaptive unit 3 · WMA12-01_2022_Oct_Q04

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6. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.
Using the laws of logarithms, solve

$$\log_4(12 - 2x) = 2 + 2\log_4(x + 1) \tag{5}$$

Adaptive unit 1 · WMA12-01_2024_Oct_Q06

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TOPIC 09 | FAMILY 04

Log Equations

WHAT THIS FAMILY ASKS

Recognise the log equations demand and complete the required reasoning in a clear, exam-ready order.

VARIANT ROUTE

V01 core family move

V02 Quadratics from logs can create invalid negative arguments.

15 classified questions

114 total marks

3. (i) Solve

$$7^{x+2} = 3$$

giving your answer in the form $x = \log_7 a$ where a is a rational number in its simplest form.

(3)

Adaptive unit 1 · WMA12-01_2021_Jan_Q03

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(ii) Using the laws of logarithms, solve

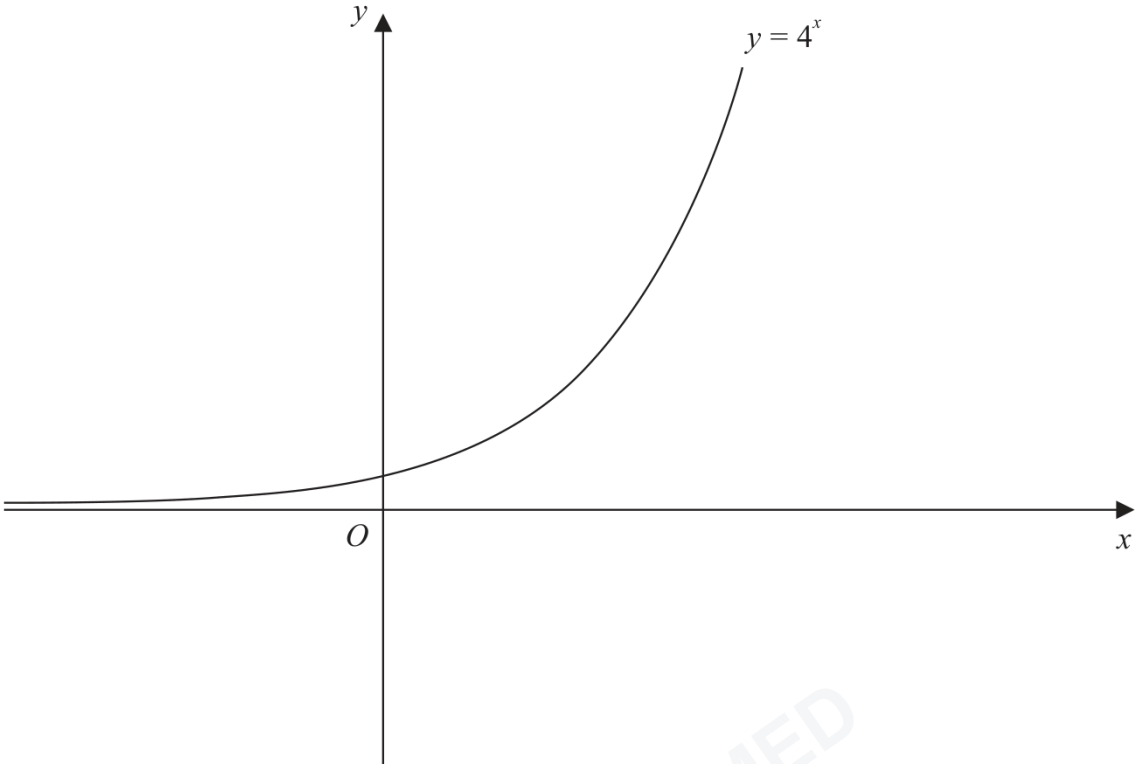
$$1 + \log_2 y + \log_2 (y + 4) = \log_2 (5 - y)$$

(5)

Adaptive unit 2 · WMA12-01_2021_Jan_Q03

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2.



Adaptive unit 1 · WMA12-01_2021_MayJune_Q02

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Figure 1

Figure 1 shows a sketch of the curve with equation $y = 4^x$

A copy of Figure 1, labelled Diagram 1, is shown on the next page.

(a) On Diagram 1, sketch the curve with equation

(i) $y = 2^x$

(ii) $y = 4^x - 6$

Label clearly the coordinates of any points of intersection with the coordinate axes.

(4)

Adaptive unit 1 · WMA12-01_2021_MayJune_Q02

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2.

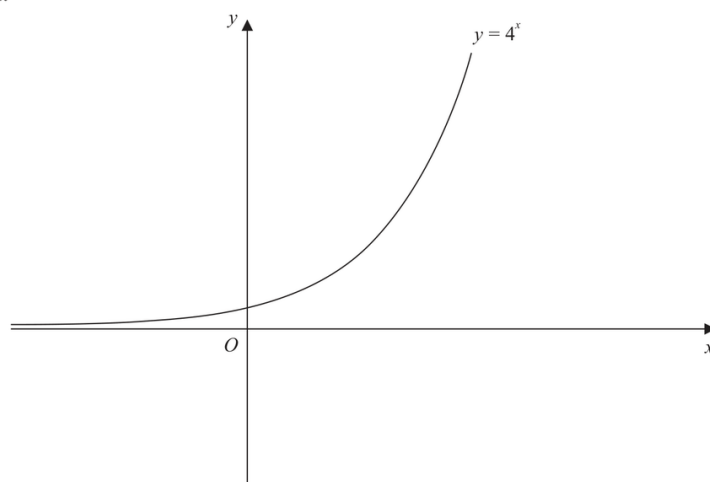
**Figure 1**

Figure 1 shows a sketch of the curve with equation $y = 4^x$

A copy of Figure 1, labelled Diagram 1, is shown on the next page.

The curve with equation $y = 2^x$ meets the curve with equation $y = 4^x - 6$ at the point P .

(b) Using algebra, find the exact coordinates of P .

(4)

Adaptive unit 2 · WMA12-01_2021_MayJune_Q02

4. Using the laws of logarithms, solve

$$\log_3(32 - 12x) = 2 \log_3(1 - x) + 3$$

(5)

Adaptive unit 1 · WMA12-01_2022_Jan_Q04

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4. In this question you must show all stages of your working.

Give your answers in fully simplified surd form.

Given that a and b are positive constants, solve the simultaneous equations

$$\begin{aligned} a - b &= 8 \\ \log_4 a + \log_4 b &= 3 \end{aligned}$$

(6)

Adaptive unit 1 · WMA12-01_2022_MayJune_Q04

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4. (i) Using the laws of logarithms, solve

$$\log_3(4x) + 2 = \log_3(5x + 7)$$

(3)

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(ii) Given that

$$\sum_{r=1}^2 \log_a(y^r) = \sum_{r=1}^2 (\log_a y)^r \quad y > 1, a > 1, y \neq a$$

find y in terms of a , giving your answer in simplest form.

(3)

Adaptive unit 2 · WMA12-01_2023_Jan_Q04

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5. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

(i) Solve

$$3^a = 70$$

giving the answer to 3 decimal places.

(2)

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5. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(ii) Find the exact value of b such that

$$4 + 3 \log_3 b = \log_3 5b \tag{4}$$

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6. (a) Given that

$$2\log_4(x + 3) + \log_4 x = \log_4(4x + 2) + \frac{1}{2}$$

show that

$$x^3 + 6x^2 + x - 4 = 0$$

(4)

Adaptive unit 1 · WMA12-01_2024_Jan_Q06

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(b) Given also that -1 is a root of the equation

$$x^3 + 6x^2 + x - 4 = 0$$

(i) use algebra to find the other two roots of the equation.

(3)

(ii) Hence solve

$$2\log_4(x+3) + \log_4 x = \log_4(4x+2) + \frac{1}{2}$$

(1)

Adaptive unit 2 · WMA12-01_2024_Jan_Q06

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9. Given that

$$3\log_2(t + 4) - 2\log_2(t - 2) = 7$$

(a) verify that $t = 4$ is a solution of the above equation,

(2)

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9. Given that

$$3 \log_2(t + 4) - 2 \log_2(t - 2) = 7$$

(b) show that

$$t^3 - 116t^2 + 560t - 448 = 0$$

(3)

Adaptive unit 2 · WMA12-01_2024_MayJuneR_Q09

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9. Given that

$$3 \log_2(t + 4) - 2 \log_2(t - 2) = 7$$

(c) Hence, using algebra and showing your working, solve

$$3 \log_2(t + 4) - 2 \log_2(t - 2) = 7$$

giving each answer in simplest form.

(Solutions based entirely on calculator technology are not acceptable.)

(4)

Adaptive unit 3 · WMA12-01_2024_MayJuneR_Q09

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7. **In this question you must show all stages of your working.**
Solutions relying entirely on calculator technology are not acceptable.

- (i) The table below shows values of x and y , where $y = \log_{10}(x + 5)$, for x values between -1 and 4

x	-1	0	1	2	3	4
$y = \log_{10}(x + 5)$	$\log_{10} 4$	$\log_{10} 5$	$\log_{10} 6$	$\log_{10} 7$	$\log_{10} 8$	$\log_{10} 9$

Using the trapezium rule with all the y values in the given table, show that

$$\int_{-1}^4 \log_{10}(x + 5) \, dx \approx \log_{10} k$$

where k is an integer to be found.

(3)

7. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(ii) Find the value of a such that

$$2\log_5(5 - a) - \log_5(a + 25) = 1 \tag{5}$$

Adaptive unit 2 · WMA12-01_2025_Jan_Q07

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In this question you must show all stages of your working. Solutions relying on calculator technology are not acceptable.

$$2\log_2(4-x) = 3 + \log_2\left(\frac{x+11}{2}\right)$$

Adaptive unit 1 · WMA12-01_2025_MayJune_Q06

6:

In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

(ii) The curves C_1 and C_2 with equations

$$y = 3^{2x+1} \quad \text{and} \quad y = 6 \times 3^x$$

meet at the point P .

Find the exact coordinates of P , writing your answer in the form $(\log_3 a, b)$ where a and b are integers.

(5)

Adaptive unit 2 · WMA12-01_2025_MayJune_Q06

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9. **In this question you must show detailed reasoning.
Solutions relying on calculator technology are not acceptable.**

(i) Solve

$$2\log_3(4x + 5) - \log_3(x + 3) = 2$$

(5)

Adaptive unit 1 · WMA12-01_2025_Oct_Q09

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9. In this question you must show detailed reasoning.
Solutions relying on calculator technology are not acceptable.

(ii) Given that $a > 0, b > 0$ and

$$\log_{10} a + \log_{10} b = \log_{10}(a + b)$$

(a) prove that $a = \frac{b}{b-1}$ (3)

(b) Hence write down the full restriction on the value of b , giving a reason for your answer. (2)

Adaptive unit 2 · WMA12-01_2025_Oct_Q09

Handwritten area with horizontal lines and a diagonal watermark reading "DR ESAM AHMED".

7. (a) Given that

$$3 \log_3(2x - 1) = 2 + \log_3(14x - 25)$$

show that

$$2x^3 - 3x^2 - 30x + 56 = 0$$

(4)

Adaptive unit 1 · WMA12-01_2021_MayJune_Q07

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(2)

(c) Hence, using algebra and showing each step of your working, solve

$$3 \log_3(2x - 1) = 2 + \log_3(14x - 25)$$

(4)

Adaptive unit 3 · WMA12-01_2021_MayJune_Q07

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5. Use the laws of logarithms to solve

$$\log_2(16x) + \log_2(x + 1) = 3 + \log_2(x + 6)$$

(5)

Adaptive unit 1 · WMA12-01_2023_MayJune_Q05

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3. **In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

(i) Using the laws of logarithms, solve

$$2\log_2(2 - x) = 4 + \log_2(x + 10)$$

(5)

Adaptive unit 1 · WMA12-01_2024_MayJune_Q03

DR ESAM AHMED

3. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(ii) Find the value of

$$\log_{\sqrt{a}} a^6$$

where a is a positive constant greater than 1

(1)

Adaptive unit 2 · WMA12-01_2024_MayJune_Q03

DR ESAM AHMED

7. **In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

(i) Find the exact solution of the equation

$$8^{2x-5} = 20$$

giving the answer in the form $a + b \log_2 5$ where a and b are rational constants.

(4)

Adaptive unit 1 · WMA12-01_2026_Jan_Q07

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7. In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

(ii) Using the laws of logarithms, solve the equation

$$\log_3(13 + 2y) + 3 = 2\log_3(4 - y)$$

(5)

Adaptive unit 2 · WMA12-01_2026_Jan_Q07

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